उप अभियंता (यांत्रिकी), महाराष्ट्र अभियांत्रिकी सेवा, गट-अ
जलसंपदा विभाग
Deputy Engineer (Mechanical), Maharashtra Engineering Service, Group-A
Water Resources Department

परीक्षेचे टप्पे:- १) लेखी परीक्षा- २०० गुण २) मुलाखत - ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विषयाशी / विभागाशी संबंधित घटक (१११९)	इंग्रजी	१००	२००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

-: अभ्यासक्रम :-

विषयाशी/विभागाशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

Sr. No.		Topics and sub-topics
1		**Current affairs (Indian and Global)** Scientific, Political, Industrial, Economic, Social, Educational, Geographical, Astronomical, Cultural etc.
2		Right To Information Act-2005, Maharashtra Right To Services Act-2015, Various Labour Laws and acts like Industrial dispute Act, Factory Act, Industrial Safety Act, Minimum Wages Act etc.
3		**General Engineering**
	a	**Engineering Materials -** Basic Crystallography, Alloys and Phase diagrams, Heat Treatment, Ferrous and Non Ferrous Metals, Non metallic materials, Basics of Nano-materials, Mechanical Properties and Testing, Corrosion prevention and control
	b	**Manufacturing, Industrial and Maintenance Engineering -** Metal casting-Metal forming, Metal Joining, Machining and machine tool operations, Limits, fits and tolerances, Metrology and inspection, computer Integrated manufacturing, FMS, Production planning and Control, Inventory control and operations research - CPM-PERT. Failure concepts and characteristics-Reliability, Failure analysis, Machine Vibration, Data acquisition, Fault Detection, Vibration Monitoring, Field Balancing of Rotors, Noise Monitoring, Wear and Debris Analysis, Signature Analysis, NDT Techniques in Condition Monitoring
	c	**IC Engines, Refrigeration and Air conditioning -** SI and CI Engines, Engine Systems and Components, Performance characteristics and testing of IC Engines; Fuels; Emissions and Emission Control. Vapour compression refrigeration, Refrigerants and Working cycles, Compressors, Condensers, Evaporators and Expansion devices, Other types of refrigeration systems like Vapour Absorption, Vapour jet, thermo electric and Vortex tube refrigeration. Psychometric properties and processes, Comfort chart, Comfort and industrial air conditioning, Load calculations and Heat pumps

	d	**Applied Thermodynamics -** Thermodynamic systems and processes, properties of pure substance, Zeroth, First and Second Laws of Thermodynamics; Entropy, Irreversibility and availability, analysis of thermodynamic cycles related to energy conversion: Rankine, Otto, Diesel and Dual Cycle, ideal and real gases, compressibility factor, Gas mixtures. Modes of heat transfer, Steady and unsteady heat conduction, Thermal resistance, Fins, Free and forced convection, Correlations for convective heat transfer, Radiative heat transfer Radiation heat transfer co-efficient, boiling and condensation, Heat exchanger performance analysis
	e	**Turbo Machinery -** Reciprocating and Rotary pumps, Pelton wheel, Kaplan and Francis Turbines, velocity diagrams, Impulse and Reaction principles, Steam and Gas Turbines, Reciprocating and Rotary Compressors Theory and Applications
	f	**Fluid Mechanics -** Basic Concepts and Properties of Fluids, Manometry, Fluid Statics, Buoyancy, Equations of Motion, Bernoulli's equation and applications, Viscous flow of incompressible fluids, Laminar and Turbulent flows, Flow through pipes and head losses in pipes. Cavitation
	g	**Engineering Mechanics -** Analysis of System of Forces, Friction, Centroid and Centre of Gravity, Dynamics; Stresses and Strains-Compound Stresses and Strains, Bending Moment and Shear Force Diagrams, Theory of Bending Stresses- Slope and deflection-Torsion, Thin and thick Cylinders, Spheres
	h	**Mechanisms and Machines -** Types of Kinematics Pair, Mobility, Inversions, Kinematic Analysis, Velocity and Acceleration Analysis of Planar Mechanisms, CAMs with uniform acceleration and retardation, cycloidal motion, oscillating followers; Vibrations Free and forced vibration of undamped and damped SDOF systems, Transmissibility Ratio, Vibration Isolation, Critical Speed of Shafts. Gears Geometry of tooth profiles, Law of gearing, Involute profile, Interference, Helical, Spiral and Worm Gears, Gear Trains- Simple, compound and Epicyclic; Dynamic Analysis Slider crank mechanisms, turning moment computations, balancing of Revolving & Reciprocating masses, Gyroscopes Effect of Gyroscopic couple on automobiles, ships and aircrafts, Governors
	i	**Design of Machine Elements-** Design for static and dynamic loading; failure theories; fatigue strength and the S-N diagram; principles of the design of machine elements such as riveted, welded and bolted joints. Shafts, Spur gears, rolling and sliding contact bearings, Brakes and clutches, flywheels
	j	**Renewable Sources of Energy -** Solar Radiation, Solar Thermal Energy collection, Solar Thermal Energy Storage, Applications - heating, cooling and Power Generation; Solar Photovoltaic Conversion; Harnessing of Wind Energy, Bio-mass and Tidal Energy - Methods and Applications, Working principles of Fuel Cells.
4		**Construction Machinery**
	a	Details of construction machineries used for construction of Earthen Dams, M.I. Tanks Canal cleaning and various types of Irrigation Tank cleaning and rebuilding works.
	b	Construction, Capacity, Utility and Use of allied equipments of construction machinery.
	c	Calculation of requirement of construction machineries for the specific work.
	d	Operation, Maintenance and Repairs of all types of construction machinery.
	e	Details of the fuels, lubricants, tyres, batteries and other allied spare parts required for construction machineries.
5		**Hydraulic Gates and Hoists -**
	a	Purpose and types of various Hydraulic Gates, Hoists and Cranes used in Irrigation Dams, Hydro Power Houses, Lift Irrigation Schemes.
	b	Construction details of the various types of Hydraulic Gates, Hoists and Cranes.
	c	Design of the all types of Hoists and Cranes required for the operation of Hydraulic gates.
	d	Details of various materials and equipments used in Manufacturing of Hydraulic Gates, Hoists and Cranes
	e	Manufacturing, Erection Testing and Commissioning of the Hydraulic Gates, Hoists and Cranes.
	f	Operation, Maintenance and Repairs of the Hydraulic Gates, Hoists and Cranes
	g	Inspection and Testing of the Hydraulic Gates, Hoists and Cranes during manufacturing, erection and commissioning
	h	Types of various operation troubles and remedies during the operation of Hydraulic Gates, Hoists and Cranes

6		**Lift Irrigation Schemes -**
	a	Layout and construction of the Lift Irrigation Scheme.
	b	Types of pumps and motors along with their construction details operation, maintenance repairs and testing used in the L.I.S
	c	Selection of type of pump with its capacity for the specific requirement of any L.I.S
	d	Types of valves along with their construction details operation, maintenance repairs used in L.I.S. and Dams
	e	Various types of allied equipments such as starters, Air Vessels, Manifolds, reducers etc. used in the L.I.S
	f	Details of various tests carried out for the various equipments used in L.I.S
	g	Testing and Commissioning of L.I.S
	h	Various type trouble shootings and remedies for that during the operation of L.I.S
	i	Various phenomena such as Cavitations, Water hammer Priming etc. in L.I.S.
7		**Road Construction Machinery and Boring**
	a	Types of Road Construction Machineries used for Construction of National and State Highway and Rural Roads
	b	Construction details Capacity, utility of the allied equipments and Road Construction Machinery`
	c	Operation, maintenance and repairs of all types of Road Construction Machinery
	d	Details of the Fuels, Lubricants, tyres, batteries and other allied spares or parts required for Road Construction Machinery
	e	Purpose of boring in the field of various projects
	f	Types of Boring machines, their construction and capacities
	g	Operation, Maintenance and Repairs of Boring Machines
	h	Various types of Boring techniques such as NX, BX Core etc

दिनांक – ११ नोव्हेंबर, २०२४

अवर सचिव
महाराष्ट्र लोकसेवा आयोग